

Lema espacioeranza condicionada mejor aprox

Lema 1. Sea $X \in \mathcal{L}^1(\mathbb{P})$ en $(\Omega, \Sigma, \mathbb{P})$ y $\mathcal{F} \subset \Sigma$ una σ -álgebra

$$\implies \forall Z \text{ } \mathcal{F}\text{-medible} : \mathbb{E}[|X - Z|^2] \geq \mathbb{E}[|X - \mathbb{E}[X | \mathcal{F}]|^2] \geq 0.$$

Demostración: Sumamos y restamos $\mathbb{E}[X | \mathcal{F}]$ dentro de $|X - Z|^2$:

$$\begin{aligned} \mathbb{E}[|X - Z|^2] &= \mathbb{E}[|X - \mathbb{E}[X | \mathcal{F}] + \mathbb{E}[X | \mathcal{F}] - Z|^2] \\ &= \mathbb{E}[|X - \mathbb{E}[X | \mathcal{F}]|^2] + \mathbb{E}[|Z - \mathbb{E}[X | \mathcal{F}]|^2] \\ &\quad + 2\mathbb{E}[(X - \mathbb{E}[X | \mathcal{F}])(\mathbb{E}[X | \mathcal{F}] - Z)] \end{aligned}$$

Veamos que $\mathbb{E}[(X - \mathbb{E}[X | \mathcal{F}])(\mathbb{E}[X | \mathcal{F}] - Z)] = 0$. Tenemos que

$$\begin{aligned} \mathbb{E}[(X - \mathbb{E}[X | \mathcal{F}])(\mathbb{E}[X | \mathcal{F}] - Z)] &= \\ &= \mathbb{E}[X \cdot Z] - \mathbb{E}[\mathbb{E}[X | \mathcal{F}] \cdot Z] + \mathbb{E}[(\mathbb{E}[X | \mathcal{F}])^2] - \mathbb{E}[X \cdot \mathbb{E}[X | \mathcal{F}]] \end{aligned} \tag{1}$$

Por definición de esperanza condicionada, $\mathbb{E}[Z \cdot \mathbb{E}[X | \mathcal{F}]] = \mathbb{E}[X \cdot Z]$ y queda 0 porque

$$\mathbb{E}\left[X \cdot \underbrace{\mathbb{E}[X | \mathcal{F}]}_{\mathcal{F}\text{-medible}}\right] = \mathbb{E}[\mathbb{E}[X | \mathcal{F}] \cdot \mathbb{E}[X | \mathcal{F}]] = \mathbb{E}[(\mathbb{E}[X | \mathcal{F}])^2].$$

Luego los dos primeros términos y los dos últimos de (??) se cancelan entre sí. ■

Observación 1. Este resultado nos dice que la esperanza condicionada es la **mejor aproximación cuadrática** de X por una variable aleatoria $\mathcal{F}$ -medible.

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